Empirical and Molecular Formulas

Answer key with worked solutions

A. Understanding

1. Empirical vs molecular formula.

> The **empirical** formula gives the simplest whole-number ratio of atoms; the **molecular** formula gives the actual number of each kind of atom in one molecule.

2. Empirical formulas.

> (a) CH (b) CH_2 (c) HO (d) C_4H_9 (e) NO_2

3. What is wrong with rounding 1 : 1.50?

> 1.50 is not close to a whole number, so it must not be rounded. Multiply both values by 2: **2 : 3**, giving A_2B_3.

B. Empirical formulas from percent composition

In a 100 g sample, each percentage becomes a mass in grams.

4. 85.63% C, 14.37% H.

> C: 85.63 g ÷ 12.01 g/mol = 7.130 mol; H: 14.37 g ÷ 1.008 g/mol = 14.26 mol. Divide by 7.130 mol: C = 1.00, H = 2.00. Empirical formula CH_2.

5. 40.04% Ca, 12.00% C, 47.96% O.

> Ca: 40.04 g ÷ 40.08 g/mol = 0.9990 mol; C: 12.00 g ÷ 12.01 g/mol = 0.9992 mol; O: 47.96 g ÷ 16.00 g/mol = 2.998 mol. Divide by 0.9990 mol: 1 : 1 : 3. Empirical formula $CaCO_3$.

6. 72.36% Fe, 27.64% O.

> Fe: 72.36 g ÷ 55.85 g/mol = 1.296 mol; O: 27.64 g ÷ 16.00 g/mol = 1.728 mol. Divide by 1.296 mol: Fe = 1.000, O = 1.333. A ratio ending in .33 means multiply by 3: 3 : 4. Empirical formula Fe_3O_4.

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C. Molecular formulas

7. CH_2O, molar mass 60.05 g/mol.

> Empirical formula mass = 12.01 g/mol + 2(1.008 g/mol) + 16.00 g/mol = 30.03 g/mol. $n = 60.05\ \text{g/mol} \div 30.03\ \text{g/mol} = 2.00$. Molecular formula $C_2H_4O_2$.

8. CH, molar mass 26.04 g/mol.

> Empirical formula mass = 12.01 g/mol + 1.008 g/mol = 13.02 g/mol. $n = 26.04\ \text{g/mol} \div 13.02\ \text{g/mol} = 2.00$. Molecular formula C_2H_2.

D. Challenge

9. 30.43% N, molar mass 92.02 g/mol.

> O = 100% − 30.43% = 69.57%. In 100 g: N: 30.43 g ÷ 14.01 g/mol = 2.172 mol; O: 69.57 g ÷ 16.00 g/mol = 4.348 mol. Ratio N : O = 1 : 2.00, so the empirical formula is NO_2 (46.01 g/mol).
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> $n = 92.02\ \text{g/mol} \div 46.01\ \text{g/mol} = 2.00$, so the molecular formula is N_2O_4.

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